

Successive Relaxation Method

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$$\mathbf{Ax} = \mathbf{b}$$

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n = b_2$$

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$$|a_{ii}| \geq \sum_{j \neq i} |a_{ij}|$$

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$$|a_{11}| \geq |a_{12}| + |a_{13}| + \cdots + |a_{1n}|$$

$$\textcolor{red}{a}_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = b_1$$

$$|a_{22}| \geq |a_{21}| + |a_{23}| + \cdots + |a_{2n}|$$

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$$|a_{33}| \geq |a_{31}| + |a_{32}| + \cdots + |a_{3n}|$$

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Recurrence Relations:

$$\tilde{x}_1^{(k)} = \frac{1}{a_{11}} \cdot (b_1 - a_{12}x_2^{(k-1)} - a_{13}x_3^{(k-1)} - \cdots - a_{1n}x_n^{(k-1)}) \quad \rightarrow \quad x_1^{(k)} = \omega \cdot \tilde{x}_1^{(k)} + (1 - \omega) \cdot x_1^{(k-1)}$$

$$\tilde{x}_2^{(k)} = \frac{1}{a_{22}} \cdot (b_2 - a_{21}x_1^{(k)} - a_{23}x_3^{(k-1)} - \cdots - a_{2n}x_n^{(k-1)}) \quad \rightarrow \quad x_2^{(k)} = \omega \cdot \tilde{x}_2^{(k)} + (1 - \omega) \cdot x_2^{(k-1)}$$

$$\tilde{x}_3^{(k)} = \frac{1}{a_{33}} \cdot (b_3 - a_{31}x_1^{(k)} - a_{32}x_2^{(k)} - \cdots - a_{3n}x_n^{(k-1)}) \quad \rightarrow \quad x_3^{(k)} = \omega \cdot \tilde{x}_3^{(k)} + (1 - \omega) \cdot x_3^{(k-1)}$$

$\vdots$

$$\tilde{x}_n^{(k)} = \frac{1}{a_{nn}} \cdot (b_n - a_{n1}x_1^{(k)} - a_{n2}x_2^{(k)} - \cdots - a_{n,n-1}x_{n-1}^{(k)}) \quad \rightarrow \quad x_n^{(k)} = \omega \cdot \tilde{x}_n^{(k)} + (1 - \omega) \cdot x_n^{(k-1)}$$

$0 < \omega < 1$
Under-Relaxation

$\omega = 1$
Gauss-Seidel

$1 < \omega < 2$
Over-Relaxation

Successive Relaxation Method

Example 03-07: Find root of the following system of equations using successive relaxation method with initial guess $\mathbf{x}^{(0)} = \mathbf{1}$, weighting factor $\omega = 1.2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$\begin{array}{rcccccccl} 10x_1 & - & x_2 & + & 2x_3 & & & = & 6 \\ - & x_1 & + & 11x_2 & - & x_3 & + & 3x_4 & = & 25 \\ & 2x_1 & - & x_2 & + & 10x_3 & - & x_4 & = & -11 \\ & & & 3x_2 & - & x_3 & + & 8x_4 & = & 15 \end{array}$$

Successive Relaxation Method

Example 03-07: Find root of the following system of equations using successive relaxation method with initial guess $\mathbf{x}^{(0)} = \mathbf{1}$, weighting factor $\omega = 1.2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$$\begin{array}{lcl} |10| > |-1| + |2| + |0| = 3 & & \\ |11| > |-1| + |-1| + |3| = 5 & - & \\ |10| > |2| + |-1| + |-1| = 4 & 2x_1 - x_2 + 10x_3 - x_4 = -11 & \\ |8| > |0| + |3| + |-1| = 4 & 3x_2 - x_3 + 8x_4 = 15 & \end{array}$$

$10x_1$	$-$	x_2	$+$	$2x_3$	$=$	6
x_1	$+$	$11x_2$	$-$	x_3	$+$	$3x_4 = 25$
$2x_1$	$-$	x_2	$+$	$10x_3$	$-$	$x_4 = -11$
		$3x_2$	$-$	x_3	$+$	$8x_4 = 15$

Successive Relaxation Method

Example 03-07: Find root of the following system of equations using successive relaxation method with initial guess $\mathbf{x}^{(0)} = \mathbf{1}$, weighting factor $\omega = 1.2$ and $\varepsilon_t = 10^{-6}$ for relative error.

$\mathbf{A}$ is diagonally dominant matrix.

$$\begin{array}{rcccccccl} & 10x_1 & - & x_2 & + & 2x_3 & & = & 6 \\ - & x_1 & + & 11x_2 & - & x_3 & + & 3x_4 & = & 25 \\ & 2x_1 & - & x_2 & + & 10x_3 & - & x_4 & = & -11 \\ & & & 3x_2 & - & x_3 & + & 8x_4 & = & 15 \end{array}$$

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Recurrence Relations:

Successive Relaxation Method

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Recurrence Relations:

$$\tilde{x}_1^{(k)} = \frac{1}{10} \cdot [6 - (-x_2^{(k-1)}) - (2x_3^{(k-1)})]$$

Successive Relaxation Method

Example 03-07: Find root of the following system of equations using successive relaxation method with initial guess $\mathbf{x}^{(0)} = \mathbf{1}$, weighting factor $\omega = 1.2$ and $\varepsilon_t = 10^{-6}$ for relative error.

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$$\begin{array}{rrrrrrrrcl} & 10x_1 & - & x_2 & + & 2x_3 & & & = & 6 \\ - & x_1 & + & 11x_2 & - & x_3 & + & 3x_4 & = & 25 \\ & 2x_1 & - & x_2 & + & 10x_3 & - & x_4 & = & -11 \\ & & & 3x_2 & - & x_3 & + & 8x_4 & = & 15 \end{array}$$

Recurrence Relations:

$$\tilde{x}_1^{(k)} = \frac{1}{10} \cdot [6 - (-x_2^{(k-1)}) - (2x_3^{(k-1)})] \quad \rightarrow \quad x_1^{(k)} = \omega \cdot \tilde{x}_1^{(k)} + (1 - \omega) \cdot x_1^{(k-1)}$$

Successive Relaxation Method

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Recurrence Relations:

$$\tilde{x}_1^{(k)} = \frac{1}{10} \cdot [6 - (-x_2^{(k-1)}) - (2x_3^{(k-1)})] \quad \rightarrow \quad x_1^{(k)} = 1.2 \cdot \tilde{x}_1^{(k)} - 0.2 \cdot x_1^{(k-1)}$$

Successive Relaxation Method

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Recurrence Relations:

$$\begin{aligned} \tilde{x}_1^{(k)} &= \frac{1}{10} \cdot [6 - (-x_2^{(k-1)}) - (2x_3^{(k-1)})] & \rightarrow & x_1^{(k)} = 1.2 \cdot \tilde{x}_1^{(k)} - 0.2 \cdot x_1^{(k-1)} \\ \tilde{x}_2^{(k)} &= \frac{1}{11} \cdot [25 - (-x_1^{(k)}) - (-x_3^{(k-1)}) - (3x_4^{(k-1)})] \end{aligned}$$

Successive Relaxation Method

Example 03-07: Find root of the following system of equations using successive relaxation method with initial guess $\mathbf{x}^{(0)} = \mathbf{1}$, weighting factor $\omega = 1.2$ and $\varepsilon_t = 10^{-6}$ for relative error.

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Recurrence Relations:

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Successive Relaxation Method

Example 03-07: Find root of the following system of equations using successive relaxation method with initial guess $\mathbf{x}^{(0)} = \mathbf{1}$, weighting factor $\omega = 1.2$ and $\varepsilon_t = 10^{-6}$ for relative error.

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Recurrence Relations:

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Recurrence Relations:

$$\begin{aligned} \tilde{x}_1^{(k)} &= \frac{1}{10} \cdot [6 - (-x_2^{(k-1)}) - (2x_3^{(k-1)})] & \rightarrow & \quad x_1^{(k)} = 1.2 \cdot \tilde{x}_1^{(k)} - 0.2 \cdot x_1^{(k-1)} \\ \tilde{x}_2^{(k)} &= \frac{1}{11} \cdot [25 - (-x_1^{(k)}) - (-x_3^{(k-1)}) - (3x_4^{(k-1)})] & \rightarrow & \quad x_2^{(k)} = 1.2 \cdot \tilde{x}_2^{(k)} - 0.2 \cdot x_2^{(k-1)} \\ \tilde{x}_3^{(k)} &= \frac{1}{10} \cdot [-11 - (2x_1^{(k)}) - (-x_2^{(k)}) - (-x_4^{(k-1)})] & \rightarrow & \quad x_3^{(k)} = 1.2 \cdot \tilde{x}_3^{(k)} - 0.2 \cdot x_3^{(k-1)} \\ \tilde{x}_4^{(k)} &= \frac{1}{8} \cdot [15 - (3x_2^{(k)}) - (-x_3^{(k)})] \end{aligned}$$

Successive Relaxation Method

Example 03-07: Find root of the following system of equations using successive relaxation method with initial guess $\mathbf{x}^{(0)} = \mathbf{1}$, weighting factor $\omega = 1.2$ and $\varepsilon_t = 10^{-6}$ for relative error.

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Recurrence Relations:

$$\begin{aligned} \tilde{x}_1^{(k)} &= \frac{1}{10} \cdot [6 - (-x_2^{(k-1)}) - (2x_3^{(k-1)})] & \rightarrow & \quad x_1^{(k)} = 1.2 \cdot \tilde{x}_1^{(k)} - 0.2 \cdot x_1^{(k-1)} \\ \tilde{x}_2^{(k)} &= \frac{1}{11} \cdot [25 - (-x_1^{(k)}) - (-x_3^{(k-1)}) - (3x_4^{(k-1)})] & \rightarrow & \quad x_2^{(k)} = 1.2 \cdot \tilde{x}_2^{(k)} - 0.2 \cdot x_2^{(k-1)} \\ \tilde{x}_3^{(k)} &= \frac{1}{10} \cdot [-11 - (2x_1^{(k)}) - (-x_2^{(k)}) - (-x_4^{(k-1)})] & \rightarrow & \quad x_3^{(k)} = 1.2 \cdot \tilde{x}_3^{(k)} - 0.2 \cdot x_3^{(k-1)} \\ \tilde{x}_4^{(k)} &= \frac{1}{8} \cdot [15 - (3x_2^{(k)}) - (-x_3^{(k)})] & \rightarrow & \quad x_4^{(k)} = 1.2 \cdot \tilde{x}_4^{(k)} - 0.2 \cdot x_4^{(k-1)} \end{aligned}$$

Successive Relaxation Method

$$k = 1$$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1,1,1,1)$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = \frac{1}{10} \cdot [6 - (-x_2^{(0)}) - 2x_3^{(0)}]$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = \frac{1}{10} \cdot [6 - (-1) - 2 \cdot 1]$$

Successive Relaxation Method

$$k = 1$$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1,1,1,1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

Successive Relaxation Method

$$k = 1$$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1,1,1,1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 1.2 \cdot \tilde{x}_1^{(1)} - 0.2 \cdot x_1^{(0)}$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 1.2 \cdot 0.5 - 0.2 \cdot 1$$

Successive Relaxation Method

$$k = 1$$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1,1,1,1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

Successive Relaxation Method

$$k = 1$$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = \frac{1}{11} \cdot \left[25 - (-x_1^{(1)}) - (-x_3^{(0)}) - 3x_4^{(0)} \right]$$

Successive Relaxation Method

$$k = 1$$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = \frac{1}{11} \cdot [25 - (-0.4) - (-1) - 3 \cdot 1]$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1,1,1,1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1,1,1,1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 1.2 \cdot \tilde{x}_2^{(1)} - 0.2 \cdot x_2^{(0)}$$

Successive Relaxation Method

$$k = 1$$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 1.2 \cdot 2.127273 - 0.2 \cdot 1$$

Successive Relaxation Method

$$k = 1$$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1,1,1,1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = \frac{1}{10} \cdot \left[-11 - 2x_1^{(1)} - (-x_2^{(1)}) - (-x_4^{(0)}) \right]$$

Successive Relaxation Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = \frac{1}{10} \cdot [-11 - 2 \cdot 0.4 - (-2.352727) - (-1)]$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1,1,1,1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

Successive Relaxation Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = 1.2 \cdot \tilde{x}_3^{(1)} - 0.2 \cdot x_3^{(0)}$$

Successive Relaxation Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = 1.2 \cdot (-0.844727) - 0.2 \cdot 1$$

Successive Relaxation Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = \frac{1}{8} \cdot [15 - 3x_2^{(1)} - (-x_3^{(1)})]$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = \frac{1}{8} \cdot [15 - 3 \cdot 2.352727 - (-(-1.213673))]$$

Successive Relaxation Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

Successive Relaxation Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 1.2 \cdot \tilde{x}_4^{(1)} - 0.2 \cdot x_4^{(0)}$$

Successive Relaxation Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 1.2 \cdot 0.841018 - 0.2 \cdot 1$$

Successive Relaxation Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2}$$

Successive Relaxation Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

Successive Relaxation Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2}$$

Successive Relaxation Method

 $k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = \frac{1}{10} \cdot [6 - (-x_2^{(1)}) - 2x_3^{(1)}]$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = \frac{1}{10} \cdot [6 - (-0.4) - 2 \cdot 2.352727]$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.2 \cdot \tilde{x}_1^{(2)} - 0.2 \cdot x_1^{(1)}$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.2 \cdot 1.078007 - 0.2 \cdot 0.4$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = \frac{1}{11} \cdot [25 - (-x_1^{(2)}) - (-x_3^{(1)}) - 3x_4^{(1)}]$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = \frac{1}{11} \cdot [25 - (-1.213609) - (-(-1.213673)) - 3 \cdot 0.809222]$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.2 \cdot \tilde{x}_2^{(2)} - 0.2 \cdot x_2^{(1)}$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.2 \cdot 2.052025 - 0.2 \cdot 2.352727$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.991884$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.991884$$

$$\tilde{x}_3^{(2)} = \frac{1}{10} \cdot \left[-11 - 2x_1^{(2)} - (-x_2^{(2)}) - (-x_4^{(1)}) \right]$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.991884$$

$$\tilde{x}_3^{(2)} = \frac{1}{10} \cdot [-11 - 2 \cdot 1.123609 - (-1.991884) - (-0.809222)]$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.991884$$

$$\tilde{x}_3^{(2)} = -1.062611$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.991884$$

$$\tilde{x}_3^{(2)} = -1.062611$$

$$x_3^{(2)} = 1.2 \cdot \tilde{x}_3^{(2)} - 0.2 \cdot x_3^{(1)}$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.991884$$

$$\tilde{x}_3^{(2)} = -1.062611$$

$$x_3^{(2)} = 1.2 \cdot (-1.062611) - 0.2 \cdot (-1.213673)$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.991884$$

$$\tilde{x}_3^{(2)} = -1.062611$$

$$x_3^{(2)} = -1.032399$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.991884$$

$$\tilde{x}_3^{(2)} = -1.062611$$

$$x_3^{(2)} = -1.032399$$

$$\tilde{x}_4^{(2)} = \frac{1}{8} \cdot [15 - 3x_2^{(2)} - (-x_3^{(2)})]$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.991884$$

$$\tilde{x}_3^{(2)} = -1.062611$$

$$x_3^{(2)} = -1.032399$$

$$\tilde{x}_4^{(2)} = \frac{1}{8} \cdot [15 - 3 \cdot 1.991884 - (-(-1.032399))] = 1.078007$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.991884$$

$$\tilde{x}_3^{(2)} = -1.062611$$

$$x_3^{(2)} = -1.032399$$

$$\tilde{x}_4^{(2)} = 0.998994$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.991884$$

$$\tilde{x}_3^{(2)} = -1.062611$$

$$x_3^{(2)} = -1.032399$$

$$\tilde{x}_4^{(2)} = 0.998994$$

$$x_4^{(2)} = 1.2 \cdot \tilde{x}_4^{(2)} - 0.2 \cdot x_4^{(1)}$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.991884$$

$$\tilde{x}_3^{(2)} = -1.062611$$

$$x_3^{(2)} = -1.032399$$

$$\tilde{x}_4^{(2)} = 0.998994$$

$$x_4^{(2)} = 1.2 \cdot 0.998994 - 0.2 \cdot 0.809222$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.991884$$

$$\tilde{x}_3^{(2)} = -1.062611$$

$$x_3^{(2)} = -1.032399$$

$$\tilde{x}_4^{(2)} = 0.998994$$

$$x_4^{(2)} = 1.036948$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.991884$$

$$\tilde{x}_3^{(2)} = -1.062611$$

$$x_3^{(2)} = -1.032399$$

$$\tilde{x}_4^{(2)} = 0.998994$$

$$x_4^{(2)} = 1.036948$$

$$\|\mathbf{x}^{(2)}\|_2 = \sqrt{(1.213609)^2 + (1.991884)^2 + (-1.032399)^2 + (1.036948)^2}$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.991884$$

$$\tilde{x}_3^{(2)} = -1.062611$$

$$x_3^{(2)} = -1.032399$$

$$\tilde{x}_4^{(2)} = 0.998994$$

$$x_4^{(2)} = 1.036948$$

$$\|\mathbf{x}^{(2)}\|_2 = \sqrt{(1.213609)^2 + (1.991884)^2 + (-1.032399)^2 + (1.036948)^2} = 2.753463$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.991884$$

$$\tilde{x}_3^{(2)} = -1.062611$$

$$x_3^{(2)} = -1.032399$$

$$\tilde{x}_4^{(2)} = 0.998994$$

$$x_4^{(2)} = 1.036948$$

$$\|\mathbf{x}^{(2)}\|_2 = \sqrt{(1.213609)^2 + (1.991884)^2 + (-1.032399)^2 + (1.036948)^2} = 2.753463$$

$$\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|_2 = \sqrt{(0.813609)^2 + (-0.360843)^2 + (0.181274)^2 + (0.227726)^2}$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

$$x_4^{(1)} = 0.809222$$

$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

$$x_2^{(2)} = 1.991884$$

$$\tilde{x}_3^{(2)} = -1.062611$$

$$x_3^{(2)} = -1.032399$$

$$\tilde{x}_4^{(2)} = 0.998994$$

$$x_4^{(2)} = 1.036948$$

$$\|\mathbf{x}^{(2)}\|_2 = \sqrt{(1.213609)^2 + (1.991884)^2 + (-1.032399)^2 + (1.036948)^2} = 2.753463$$

$$\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|_2 = \sqrt{(0.813609)^2 + (-0.360843)^2 + (0.181274)^2 + (0.227726)^2} = 0.936422$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1,1,1,1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

$$\tilde{x}_2^{(1)} = 2.127273$$

$$x_2^{(1)} = 2.352727$$

$$\tilde{x}_3^{(1)} = -0.844727$$

$$x_3^{(1)} = -1.213673$$

$$\tilde{x}_4^{(1)} = 0.841018$$

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$$\|\mathbf{x}^{(1)}\|_2 = \sqrt{(0.4)^2 + (2.352727)^2 + (-1.213673)^2 + (0.809222)^2} = 2.796993$$

$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

$$\tilde{x}_1^{(2)} = 1.078007$$

$$x_1^{(2)} = 1.213609$$

$$\tilde{x}_2^{(2)} = 2.052025$$

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$$\tilde{x}_3^{(2)} = -1.062611$$

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$$\|\mathbf{x}^{(2)}\|_2 = \sqrt{(1.213609)^2 + (1.991884)^2 + (-1.032399)^2 + (1.036948)^2} = 2.753463$$

$$\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|_2 = \sqrt{(0.813609)^2 + (-0.360843)^2 + (0.181274)^2 + (0.227726)^2} = 0.936422$$

$$\varepsilon^{(2)} = \frac{\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|_2}{\|\mathbf{x}^{(2)}\|_2} = \frac{0.936422}{2.753463} = 0.340089 > \varepsilon_t$$

Successive Relaxation Method

$k = 1$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(0)} = (1, 1, 1, 1)$$

$$\tilde{x}_1^{(1)} = 0.5$$

$$x_1^{(1)} = 0.4$$

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$$\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2 = \sqrt{(-0.6)^2 + (1.352727)^2 + (-2.213673)^2 + (-0.190778)^2} = 2.669572$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_2}{\|\mathbf{x}^{(1)}\|_2} = \frac{2.669572}{2.796993} = 0.954444 > \varepsilon_t \rightarrow \text{continue}$$

$k = 2$

$$\mathbf{x}^{(k-1)} = \mathbf{x}^{(1)} = (0.4, 2.352727, -1.213673, 0.809222)$$

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$$\varepsilon^{(2)} = \frac{\|\mathbf{x}^{(2)} - \mathbf{x}^{(1)}\|_2}{\|\mathbf{x}^{(2)}\|_2} = \frac{0.936422}{2.753463} = 0.340089 > \varepsilon_t \rightarrow \text{continue}$$